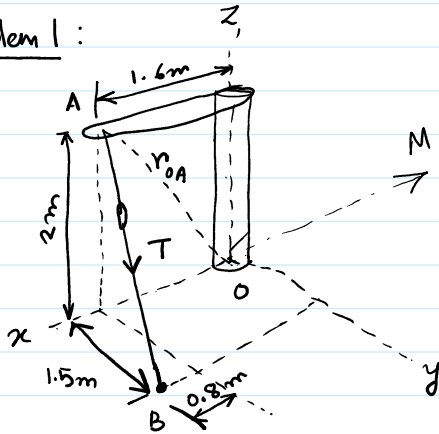


Tutorial Moment SEC D

20 January 2022 09:54

Problem 1:



The turnbuckle is tightened until tension in the cable AB is 2.4 kN. Determine the moment about pt O of the cable force acting on point A and magnitude of this moment.

Solution: $O(0,0,0)$
 $A(1.6, 0, 2)$
 $B(2.4, 1.5, 0)$

$$\vec{AB} = 0.8\hat{i} + 1.5\hat{j} - 2\hat{k}$$

$$\vec{n}_{AB} = \frac{0.8\hat{i} + 1.5\hat{j} - 2\hat{k}}{\sqrt{(0.8)^2 + (1.5)^2 + (-2)^2}}$$

$$\vec{T} = 2.4 \vec{n}_{AB} \text{ kN}$$

$$= 0.731\hat{i} + 1.371\hat{j} - 1.829\hat{k} \text{ kN}$$

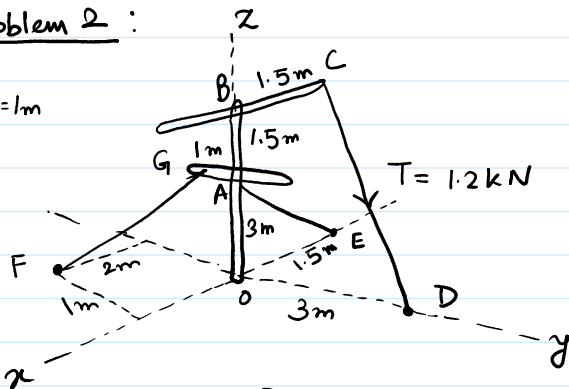
$$\vec{r} = 1.6\hat{i} + 2\hat{k}$$

$$M_O = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1.6 & 0 & 2 \\ 0.731 & 1.371 & -1.829 \end{vmatrix} \text{ kN-m}$$

From the moment vector find magnitude. (5.62 kN-m)

Problem 2:

$AG = 1\text{m}$



This is a rigid pole and cross-arm assembly. Determine the vector expression for moment of 1.2 kN tension about point O and about the pole z-axis.

$$O(0,0,0)$$

$$C(-1.5, 0, 4.5)$$

$$D(0, 3, 0)$$

$$\vec{CD} = 1.5\hat{i} + 3\hat{j} - 4.5\hat{k}$$

$$\vec{n}_{CD} = \frac{1.5\hat{i} + 3\hat{j} - 4.5\hat{k}}{\sqrt{1.5^2 + 3^2 + 4.5^2}}$$

$$\vec{r} = -1.5\hat{i} + 4.5\hat{k}$$

$$\vec{T} = 1.2(\vec{n}_{CD}) \quad \text{--- (Step I)}$$

$$= 0.45\hat{i} + 0.9\hat{j} - 1.35\hat{k}$$

$$M_O = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1.5 & 0 & 4.5 \\ 0.45 & 0.9 & -1.35 \end{vmatrix} \text{ kN-m} = -4.05\hat{i} - 1.35\hat{k} \text{ kN-m} \quad \text{--- (Step II)}$$

Unit vector about the pole axis is $\hat{k}$. So moment about pole axis.

$$M_z = \hat{k} \cdot (-4.05\hat{i} - 1.35\hat{k}) \text{ kN-m. (Magnitude)}$$

Multiply with $\hat{k}$ again
to get the moment
vector.

Ans: $M_o = -2.89\hat{i} - 0.962\hat{k} \text{ kN-m}$
 $M_z = -0.962\hat{k} \text{ kN-m}$